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Dear Editor,

We are pleased to submit our revised manuscript entitled “*A Kirkpatrick-based Framework for Integrating Safety and Sustainability in Aviation Maintenance: A Case Study on Waste Reduction and Resource Efficiency*”, for consideration as a Case Study in the *Journal of Air Transport Management*.

The aviation maintenance, repair and overhaul (MRO) sector sits at the intersection of two pressures that the journal regularly addresses: safety-driven regulation under ICAO/FAA/EASA/ANAC oversight, and the rising demand for credible sustainability disclosures from airlines and their supply chain. Maintenance errors contribute to between 12% and 18% of commercial aviation accidents and simultaneously generate substantial material, energy and labour waste. Despite the strategic relevance of technical training as a high-leverage intervention, empirical evidence on how to operationalise training evaluation beyond Level 2 of the Kirkpatrick model in AS9100-certified MRO settings is scarce, and the explicit translation of training-effectiveness gains into cleaner-production outcomes has not been documented in a single longitudinal case.

Our study addresses this gap through a twelve-month case study at a multinational aircraft-engine MRO facility. We operationalise the four Kirkpatrick levels (Reaction, Learning, Behaviour, Results) within the existing training process, with each level mapped onto specific AS9100 clauses, SMS pillars and sustainability indicators. The intervention was associated with a 28% lower training-failure rate and a 35% lower regulatory non-conformance rate, with estimated annual quality-cost savings of USD 0.75–1.5 M across pessimistic-to-optimistic scenarios (USD 1.19 M base case) and a payback period under twelve months in every scenario. Customer satisfaction surveys with four major airline customers reported improvements across all five evaluated categories during the intervention year. The single-site case-study design supports association rather than causal attribution, and we report the results accordingly. We derive managerial implications for MRO operators and policy recommendations for civil aviation authorities and industry bodies (IATA, SAE).

We believe this work aligns directly with the editorial scope of JATM in three respects:

- It contributes to the journal’s recurring theme of aviation safety management, providing measurable evidence of how training evaluation translates into regulatory compliance.
- It contributes to the growing JATM stream on sustainability in air transport, demonstrating an auditable connection between human capital interventions and cleaner production.
- It offers actionable policy guidance addressed to civil aviation authorities and industry bodies, in line with the journal’s managerial and policy orientation.

In response to the previous round of constructive feedback, the revised manuscript adds a dedicated *Managerial Implications* section, a *Policy Recommendations* section, an elaborated

Economic Analysis of cost-of-quality decomposition, a comparative *Before/After* results table, a structured *Limitations* table, and a new conceptual figure linking Kirkpatrick levels to AS9100 clauses and sustainability outcomes. The abstract has been rewritten to lead with economic outcomes, and the literature review has been strengthened with recent JATM references.

We confirm that:

- This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere.
- All authors have read and approved the submission.
- There are no known conflicts of interest associated with this work.
- Aggregated operational data are reported within the manuscript; individual-level data are not publicly available due to commercial confidentiality.

Should you require any additional information, please do not hesitate to contact me.

Sincerely,

Dr. José Cristiano Pereira
Corresponding author
on behalf of the authors